Western Digital.

Western Digital

MLPerf Storage 2.0 Closed
Remotely-Attached Block Device
OpenFlex Data24 4200 with PEAK:AIO AI Data Server SDS and One Client.

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Hardware

Data24_PEAK_48Drive_1Node

- Client Servers:
  - SuperMicro 521-GE-TNRT Qty 1
  - Form Factor: 5U
  - CPU: 2x Intel Xeon Platinum 8462Y
  - Memory: 1024GB (16x64GB DDR5)
  - OS: Ubuntu 22.04
  - Data Network: Nvidia CX7
- Disaggregated Storage:
  - OpenFlex Data24 4200 Qty 2
  - Form Factor: 2U
  - 48x SanDisk SN655 15.36TB
- PEAK:AIO Server
  - ASUS WS Motherboard
  - Form Factor: 4u
  - CPU: 1x Intel W9-3475X
  - Memory: 512GB (8x64GB DDR5)
  - OS: PEAK:AIO Linux 24.04
  - Data Network: Nvidia CX7
- Switch:
  - Mellanox SN3800 64-Port 100GbE QSFP28 with Onyx OS

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Software and Settings
Data24_PEAK_48Drive_1Node

OpenFlex Data24 EBOF
- NVMe Protocol: RDMA

Switch
- Congestion Control: PFC

Peak:AIO Linux Appliance
- Default settings

Client Server
- OS: Ubuntu 22.04
- Kernel: 5.15.0-56-generic
- OFED Version: 24.10-2.1.8.0-ubuntu22.04-x86_64
- Hyperthreads: disabled
- Turbo: enabled
- Sysctl settings:
  - `vm.swappiness=0`
  - `vm.dirty_bytes=50000000`
  - `dev.raid.speed_limit_max=5000000`
  - `dev.raid.speed_limit_min=50000`
  - `net.ipv4.tcp_timestamps=0`
  - `net.ipv4.tcp_sack=1`
  - `net.core.netdev_max_backlog=250000`
  - `net.core.rmem_max=268435456`
  - `net.core.wmem_max=268435456`
  - `net.core.rmem_default=4194304`
  - `net.core.wmem_default=4194304`
  - `net.core.optmem_max=4194304`
  - `net.ipv4.tcp_rmem=4096 87380 134217728`
  - `net.ipv4.tcp_wmem=4096 65536 134217728`
  - `net.ipv4.tcp_low_latency=1`
  - `net.ipv4.tcp_adv_win_scale=1`
  - `net.ipv4.tcp_mtu_probing=1`
  - `net.ipv6.conf.all.disable_ipv6=1`
  - `net.ipv6.conf.default.disable_ipv6=1`
  - `net.ipv6.conf.lo.disable_ipv6=1`
  - `fs.aio-max-nr=1048576`

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Nameplate Power

Data24_PEAK_48Drive_1Node

Category	Component	Quantity	PSU Nameplate Power	Design Power
Storage	Data24 4200	1	2 x 800W	800W
SDS Server	PEAK:AIO Server	1	1 x 1200W	1200W
			Total Power	2000W

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Nameplate Power

DLIO Clients

- Quantity: 1
- PSU1 Nameplate Power: 2700W
- PSU2 Nameplate Power: 2700W
- Design Power: 5400W

PEAK:AIO Server

- Quantity: 1
- PSU1 Nameplate Power: 1200W
- Design Power: 1200W

Storage and Network

Storage Data Node

- Quantity: 1 OpenFlex Data24 4200
- PSU Nameplate: 2 x 800W
- Design Power: 800W

Backend Switch

- Quantity: 1
- PSU Nameplate: 2 x 1100W
- Design Power: 1100W

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Topology

Data24 With PEAK:AIO

Data Storage Network

Client Server Network

